Grades will be notified at

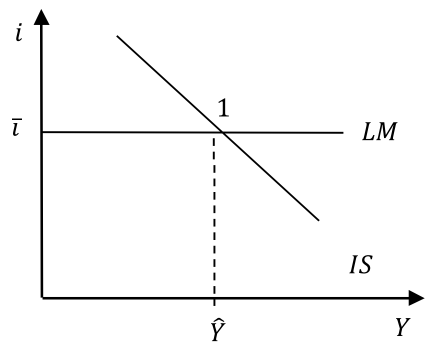
7:30pm of Tuesday, July 2, 2024

Question 1 (5 points)

Zeta is a closed economy with constant prices described by an IS-LM model. In Zeta, investiment, net taxes and government purchases of goods and services are exogenous ($I = \bar{I}$, $T = \bar{T}$ and $G = \bar{G}$), and the central bank chooses the interest rate, keeping it constant at the level $i = \bar{\imath}$ (> 0). Finally, the consumption function is

$$C = c_0 + c_1(Y - \bar{T}) - hi + a\,\overline{WFH},$$

where the parameter h (> 0) is the sensitivity of consumption to the interest rate, the parameter a (> 0) is the sensitivity of consumption to the financial and housing wealth WFH (assumed to be exogenous, so that $WFH = \overline{WFH}$), and the other symbols have the usual meaning. Represent in the graph below the initial equilibrium of Zeta, clearly indicating the variables measured along the axes, and explain the reason for the slopes of the curves you have drawn. Suppose now that individuals' financial and housing wealth falls ($\Delta\overline{WFH} < 0$) and that, to prevent this change from affecting equilibrium income, Zeta's government decides to vary, at the same time and by the same amount, net taxes and its purchases of goods and services, so that $\Delta\bar{T} = \Delta\bar{G}$. Explain if and why, to achieve its aim, the government should cut or rise $\bar{T}$ and $\bar{G}$, showing in the graph the final equilibrium that the economy will reach following the decrease in $\overline{WFH}$ and the government's intervention. In this new equilibrium, private saving will be higher, lower, or unchanged? Why? Explain, by making explicit reference to the private saving function that prevails in Zeta. *[NB: you are NOT being asked to analytically solve the IS-LM model for this economy, or to compute the analytical expression for the change in private saving, but just to provide* <u>*the economic intuition*</u> *that helps determine the way private saving will be affected by the changes under consideration.]*

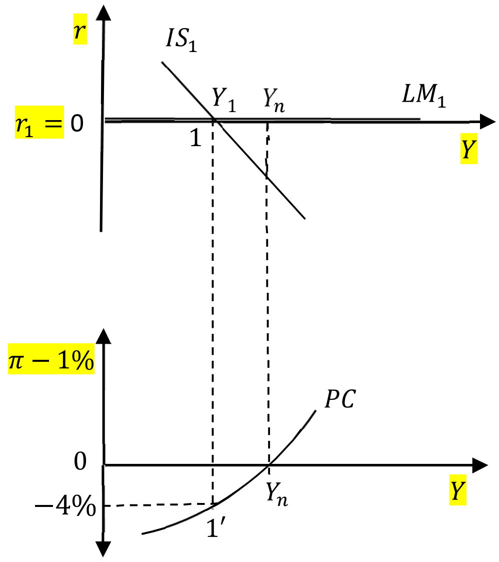


SKETCH OF ANSWER

In Zeta, the LM is horizontal at the level of the interest rate $i = \bar{\imath}$ chosen by the central bank, and the IS is negatively sloped as usual (although investment is exogenous, consumption– and therefore aggregate demand – depend negatively on i). If $\overline{WFH}$ falls, consumption decreases and the IS shifts to the left. An equal, simultaneous change in net taxes and public purchases of goods and services causes autonomous demand to vary by $\Delta A = (1 - c_1)\Delta\bar{G}$. It follows that, in order to keep equilibrium income unchanged, the government will have to increase $\bar{T}$ and $\bar{G}$ by the amount needed to bring back the IS to its original position, shifting it to the right. The equilibrium, initially point 1 in the figure, remains 1. Following the decrease in $\overline{WFH}$ and the goverment's intervention just described, private saving will be unchanged – as it must be necessarily true, being investment exogenous and government saving unchanged. To explain why individuals will not change their saving decisions is the fact that there are two forces at play: the fall in $\overline{WFH}$, that induces individuals to save more, and the increase in $\bar{T}$, that induces them to save less. In equilibrium, these two forces will balance each other exactly.

Question 2 (5 points)

An economy in which $\pi^e = 1\%$ and described by an IS-LM-PC model finds itself in the initial short-run equilibrium represented by the pair $(1,1')$ in the graph below, with $Y = Y_1 < Y_n$.
Having explained which variable is measured along the axes of each of the two graphs, write down the values that the inflation rate (π), the real interest rate (r) and the nominal interest rate (i) will take on today in the equilibrium $(1,1')$, motivating your answers. Furthermore, discuss if and why, absent interventions by the economy's policy-makers and under the same assumptions specified above, next period the real interest rate, income, inflation and the nominal interest rate will be higher, lower or unchanged compared to their current levels.



SKETCH OF ANSWER

The variable measured along the vertical axis of the IS-LM diagram is the real interest rate, while the one measured along the horizontal axis of both graphs is real GDP, Y. Finally, the variable on the vertical axis of the graph that depicts the PC curve is, as usual, the difference between current and expected inflation – that is, in this economy in which $\pi^e = 1\%$, $\pi - 1\%$. It follows that, at point 1, $r = 0\%$, while $\pi^e = 1\%$ and $i = r + \pi^e$ imply $i = 0 + 1\% = 1\%$. Finally, from the second graph it follows that, at $1'$, $\pi - 1\% = -4\%$, and therefore that $\pi = -4\% + 1\% = -3\%$. Absent interventions by the economy's policy-makers and given that individuals will continue to expect an inflation rate of 1%, next period the IS and the LM curves, the real and the nominal interest rates and income will be unchanged, so that the economy will remain in the pair of points $(1,1')$ in the figure.

Question 3 (5 points)

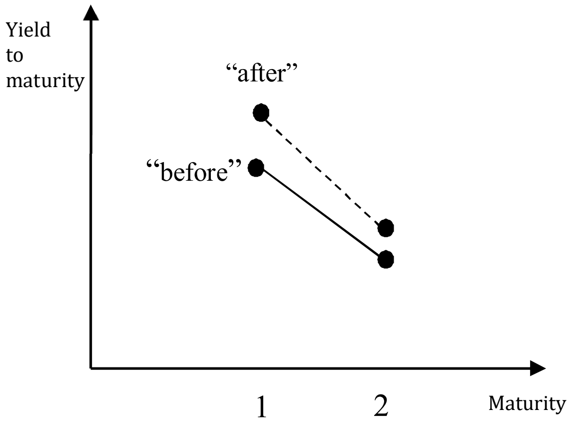
In country Alpha there are only one-year and two-year bonds. Current and expected future inflation are equal to zero, so nominal and real interest rates are the same. In period $t+1$ ("the future"), consumption depends only on disposable income at time $t+1$, while investment depends on the values that interest rate and income will assume at time $t+1$. As for the current period (t, "the present"), consumption depends positively on disposable income at time t and on the expected disposable income at time t+1, $C_t = C(Y_t - \bar{T}_t, Y^e_{t+1} - \bar{T}^e_{t+1})$, while investment depends positively on current and expected future income levels, and negatively on the current and future expected levels of short-term interest rates. In both periods, the central bank chooses the interest rate, and G and T are exogenous as usual. Finally, at time t, in the initial equilibrium position both income and the interest rate take on positive values, while at time $t+1$ the economy is expected to be in a 'liquidity trap'. Suppose now that, at time t, the government announces that at time $t+1$ it will increase its purchases of goods and services. At the same time, the central bank announces that, at time t, it will adjust the policy rate so as to prevent any change in the equilibrium level of income for time t, and time t only (Y_t). Explain how these fiscal and monetary policy announcements, which come as a surprise to individuals and are believed, will change the yield to maturity of the two-year bonds at time t, i_{2t}.

SKETCH OF ANSWER

The increase in $\bar{G}_{t+1}$ will shift the IS for that period to the right, leading to an equilibrium characterized by a higher Y_{t+1} and an interest rate $i_{1,t+1}$ still equal to zero (in fact, being the economy in a liquidity trap at time $t+1$, the intersection between the new IS and the LM curve will continue to take place for an interest rate equal to zero). The higher expected future income will raise consumption and investment at time t, thus causing a rightward shift of the IS. To prevent Y_t from rising, the central bank will have to raise the time t policy rate. Since i_{2t} is (approximately) equal to the average of the short-term current and expected future interest rates, and that $i_{1,t+1}$ remains equal to zero while $i_{1,t}$ rises, the yield to maturity of the two-year bonds at time t will certainly increase.

Question 4 (5 points)

Using the information provided in the previous question, draw the yield curve prevailing in Alpha at time t before the central bank and government's announcements, and explain how it will be affected by those announcements. In particular, draw the new yield curve in the same graph, and explain if it is possible to determine without ambiguity how its slope and position will compare to those of the original curve.

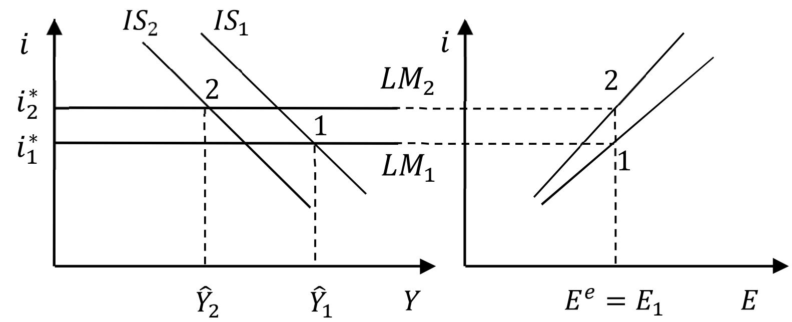


SKETCH OF ANSWER

The initial yield curve is the continuous one in the figure. Its first point is the positive value the current short rate i_{1t} *is initially taking on in this economy; being* $i_{2t} = \frac{1}{2}\left(i_{1t} + i^e_{1,t+1}\right)$ *and* $i^e_{1,t+1} = 0$, *its second point is* $i_{2t} = \frac{1}{2} i_{1t}$ $(< i_{1t})$. *The new curve (the dashed line in the figure) will lie above, and will be steeper than, the initial one. In fact, its first point will go up by* Δi_{1t} *(the amount by which the central bank will raise the policy rate at time* t*) and its second one by* $\Delta i_{2t} = \frac{1}{2}\Delta i_{1t}$, *which is less than* Δi_{1t}.

Question 5 (5 points)

Consider an open economy with constant prices under a fixed exchange rate regime, in which net income from abroad is zero ($NI = 0$) and investment is entirely exogenous ($I = \bar{I}$). Aside from the exogeneity of investment, the economy is described by a standard "IS-LM-interest parity" model: the Marshall-Lerner condition is met, consumption depends on disposable income only, and G and T are exogenous. Represent the initial equilibrium of the economy in the pair of graphs below, denoting it by '1', assuming that the foreign interest rate is initially i_1^*, the exchange rate equal to E_1, and that market participants expect that i twill be kept at E_1 also in the future, so that $E^e = E_1$.
Consider now an increase in the foreign interest rate, that becomes $i_2^* > i_1^*$, and a contemporaneous fall in foreign income, Y^*. Assuming that the central bank keeps the exchange rate fixed at E_1, explain if and how each of the curves depicted in the two graphs will be affected by these two contemporaneous changes. Represent the new equilibrium that the economy will reach, denoting it by '2' in the two graphs, and explain if and why it is, or is not, possible to conclude that, in this new equilibrium, net exports (NX) will be higher, lower, or unchanged.



SKETCH OF ANSWER

When i^* rises, in the second graph the straight line representing the interest parity condition rotates upwards and to the left; in the first one, the decrease in foreign income shifts the IS curve to the left. If the central bank kept the domestic interest rate at the initial level (i_1^*), the exchange rate would depreciate. It follows that, to keep it fixed at E_1, the central bank will have to bring the domestic interest rate at the higher level the foreign one is taking on, i_2^*. The new equilibrium becomes '2' in the figure. In the move from '1' to '2', net exports tend to rise for the decrease in domestic income, and to fall due to the decrease in foreign income. However, the latter force will prevail, so that net exports will be lower in the new equilibrium. In fact, in an open economy's goods market equilibrium, national saving is equal to the sum of investment and net exports. But in this country investment is unchanged (it is exogenous), implying that net exports must change in the same direction, and by the same amount, as national saving. Being national saving lower in the new equilibrium (government saving is unchanged; private saving has gone down because of the decrease in Y), the same must be true about net exports.

Question 6 (5 points)

At time $t = 1$, the economy of Country Beta is characterized by the following data: $r = 15\%,\ g = 10\%,\ Y_0 = 60,\ T_1 = 10,\ G_1 = 10,\ B_0 = 30.$ Compute the numerical value of Y_1 – real GDP at time 1 – for this economy and, using the relation that gives the evolution of the debt-to-GDP ratio over time, B_1/Y_1 – the value that, absent policy interventions, that ratio will take on at time 1. Finally, compute the value of the primary deficit-to-GDP ratio that, at time 1, would stabilize the debt-to-GDP ratio at the value it was taking on at time 0.

SKETCH OF ANSWER

In this economy, one can compute time 1 *GDP as* $Y_1 = (1+g)Y_0 = (1+0.1)\cdot 60 = 66.$ *Also notice that, in the same time period, the primary deficit-to-GDP ratio* (d_1) *is zero. Plugging this values and the other available numerical information on this economy in the equation that gives the dynamics of the debt ratio, yields:*

$$\begin{aligned} \frac{B_1}{Y_1} &= (1+r-g)\frac{B_0}{Y_0} \\ &= (1.05)\frac{30}{60} \\ &= 0.525 \end{aligned}$$

The value of the primary deficit-to-GDP ratio that, at time 1*, would stabilize the debt ratio is the unique* d *that solves the following equation:*

$$\frac{B_1}{Y_1} - \frac{B_0}{Y_0} = (r-g)\frac{B_0}{Y_0} + d = 0,$$

that is:

$$d = (g-r)\frac{B_0}{Y_0} = (-0.05)\cdot 0.5 = -0.025.$$

To stabilize the debt ratio in this economy, the government must run a primary surplus of 2.5% *of the country's GDP, something that amounts to transforming the time 0 debt ratio into a steady state.*